US 20250272214A1

(19) United States

(12) Patent Application Publication (10) Pub. No.: US 2025/0272214 A1

Randhi et al. (43) Pub. Date: Aug. 28, 2025

(54) **SYSTEM AND METHOD FOR GENERATING SOFTWARE CODE CRITICALITY AND EXECUTION HEATMAPS TO OPTIMIZE RESOURCE CONSUMPTION**

(71) Applicant: **BANK OF AMERICA CORPORATION**, Charlotte, NC (US)

(72) Inventors: **Venugopala Rao Randhi**, Hyderabad (IN); **Rama Venkata S. Kavali**, Frisco, TX (US); **Damodarrao Thakkalapelli**, Plano, TX (US); **Vijaya Kumar Vegulla**, Hyderabad (IN)

(73) Assignee: **BANK OF AMERICA CORPORATION**, Charlotte, NC (US)

(21) Appl. No.: **18/585,525**

(22) Filed: **Feb. 23, 2024**

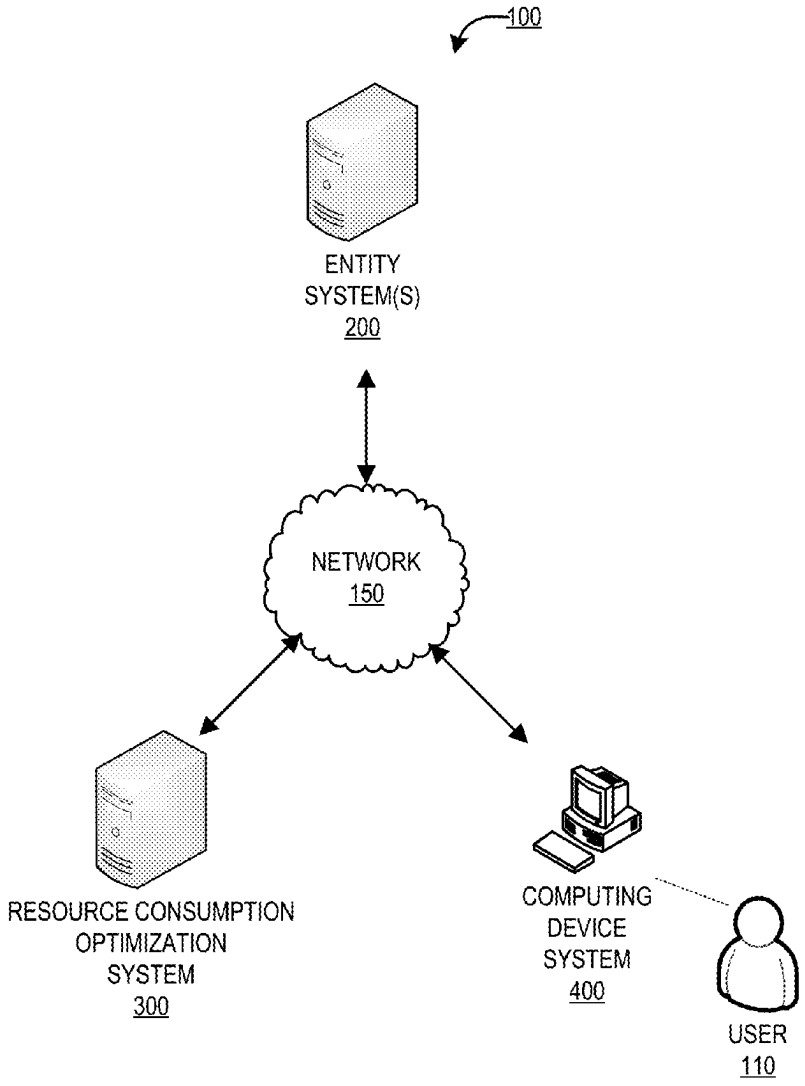
Publication Classification

(51) **Int. Cl.**
G06F 11/36 (2025.01)

(52) **U.S. Cl.**
CPC ***G06F 11/3604*** (2013.01); ***G06F 11/3676*** (2013.01)

(57) **ABSTRACT**

Embodiments of the present invention provide a system for generating software code criticality and execution heatmaps to optimize resource consumption. The system is configured for accessing one or more entity applications associated an entity, wherein the one or more entity applications comprises lines of program code, parsing through the lines of program code associated with the one or more entity applications, generating tokens for each of the lines of program code, generating heatmap associated with each of the lines of program code, wherein the heatmap comprises visual representations associated with each of the tokens generated for each of the line of program code, and displaying the heatmap to one or more users.



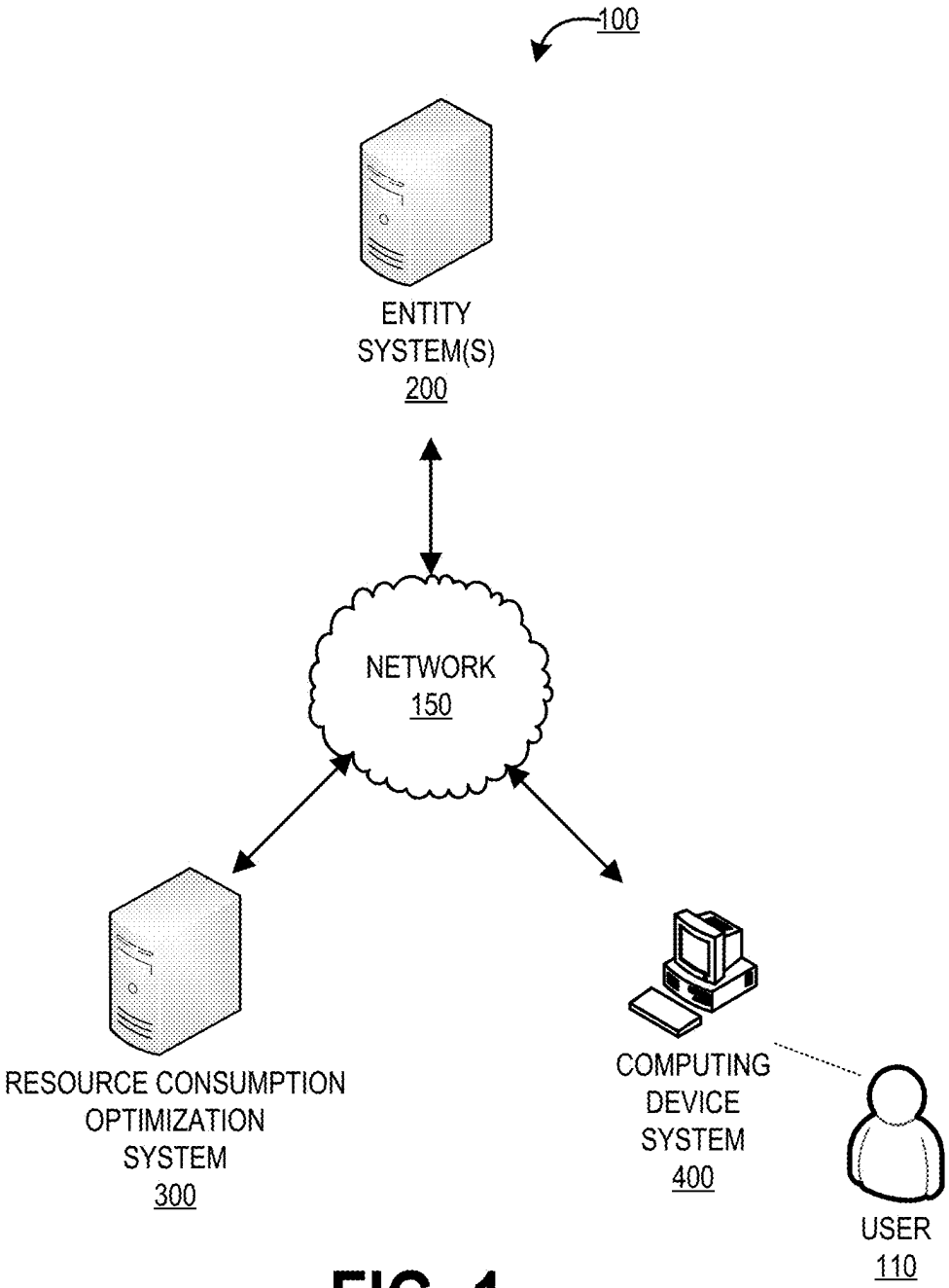

100
ENTITY SYSTEM(S) 200
NETWORK 150
RESOURCE CONSUMPTION OPTIMIZATION SYSTEM 300
COMPUTING DEVICE SYSTEM 400
USER 110

FIG. 1

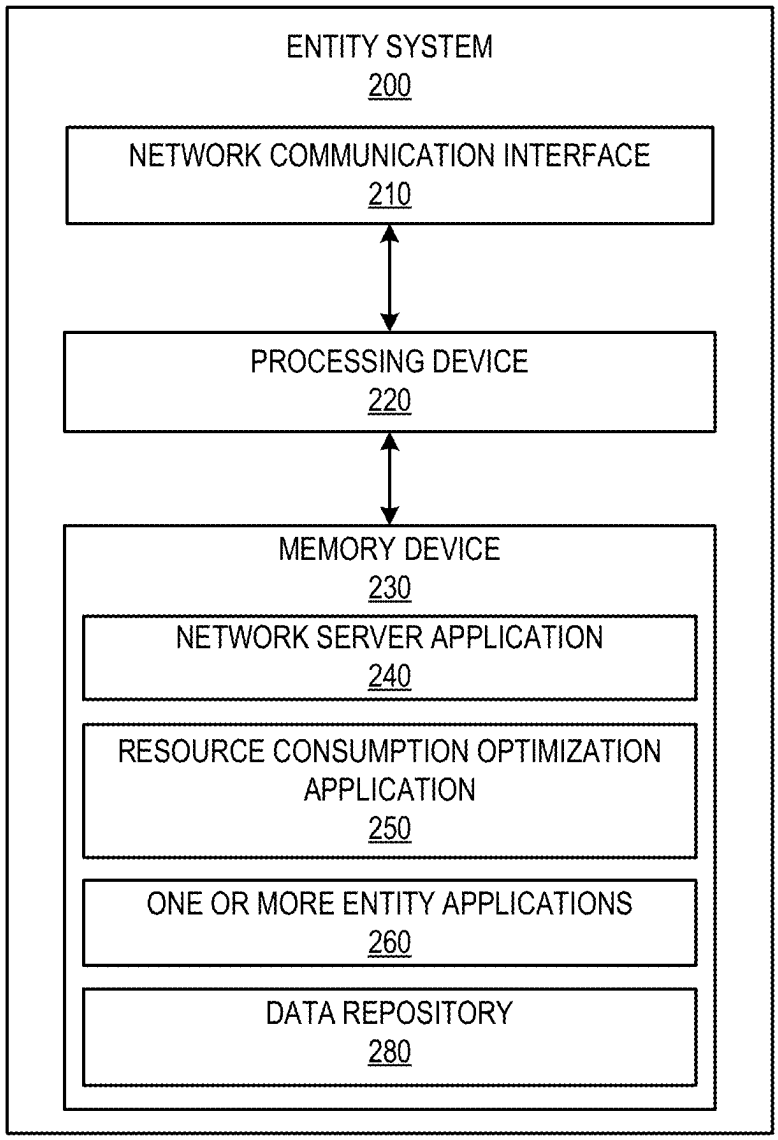

ENTITY SYSTEM
200
NETWORK COMMUNICATION INTERFACE
210
PROCESSING DEVICE
220
MEMORY DEVICE
230
NETWORK SERVER APPLICATION
240
RESOURCE CONSUMPTION OPTIMIZATION APPLICATION
250
ONE OR MORE ENTITY APPLICATIONS
260
DATA REPOSITORY
280

FIG. 2

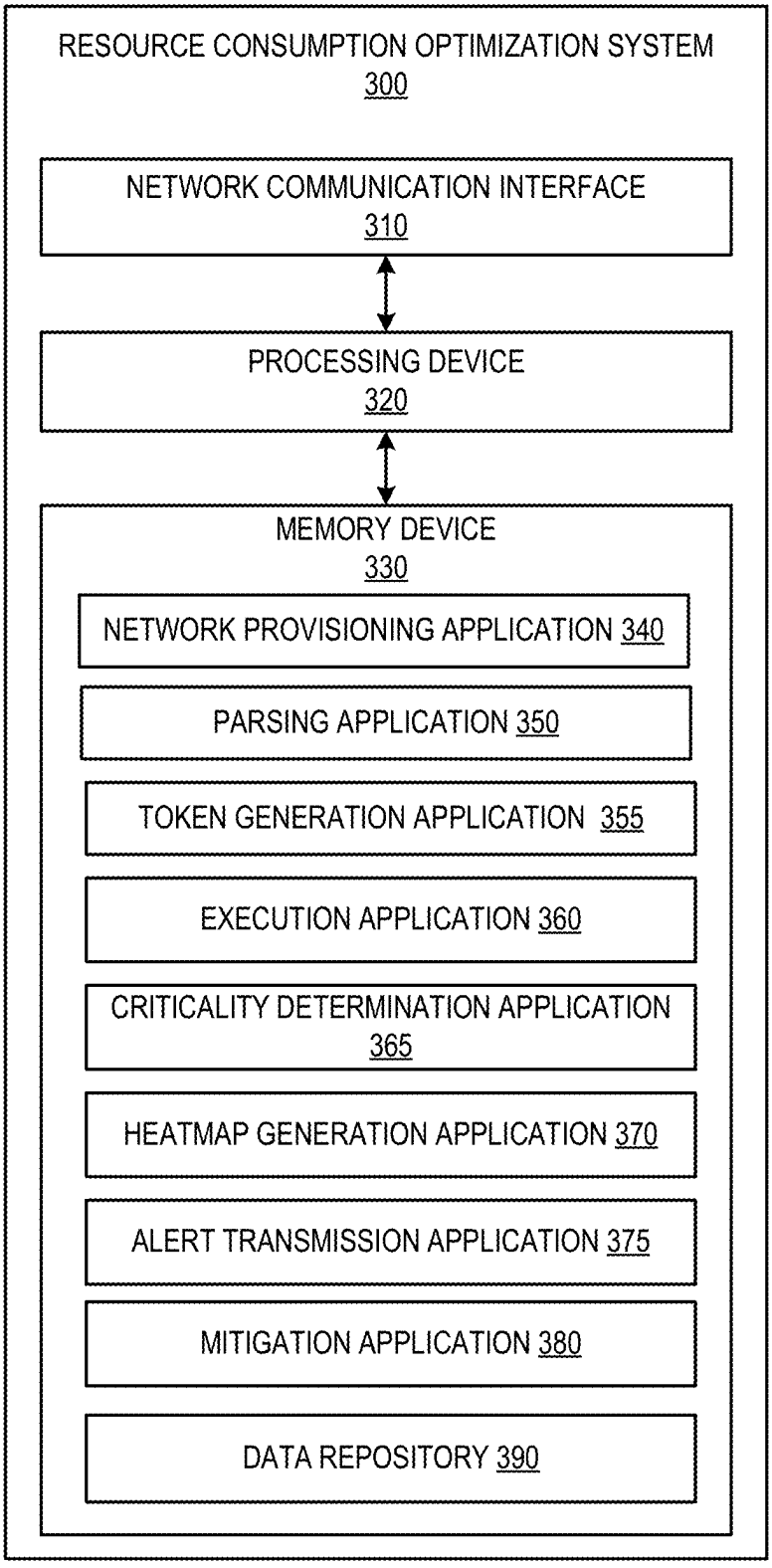

RESOURCE CONSUMPTION OPTIMIZATION SYSTEM
300
NETWORK COMMUNICATION INTERFACE
310
PROCESSING DEVICE
320
MEMORY DEVICE
330
NETWORK PROVISIONING APPLICATION 340
PARSING APPLICATION 350
TOKEN GENERATION APPLICATION 355
EXECUTION APPLICATION 360
CRITICALITY DETERMINATION APPLICATION
365
HEATMAP GENERATION APPLICATION 370
ALERT TRANSMISSION APPLICATION 375
MITIGATION APPLICATION 380
DATA REPOSITORY 390

FIG. 3

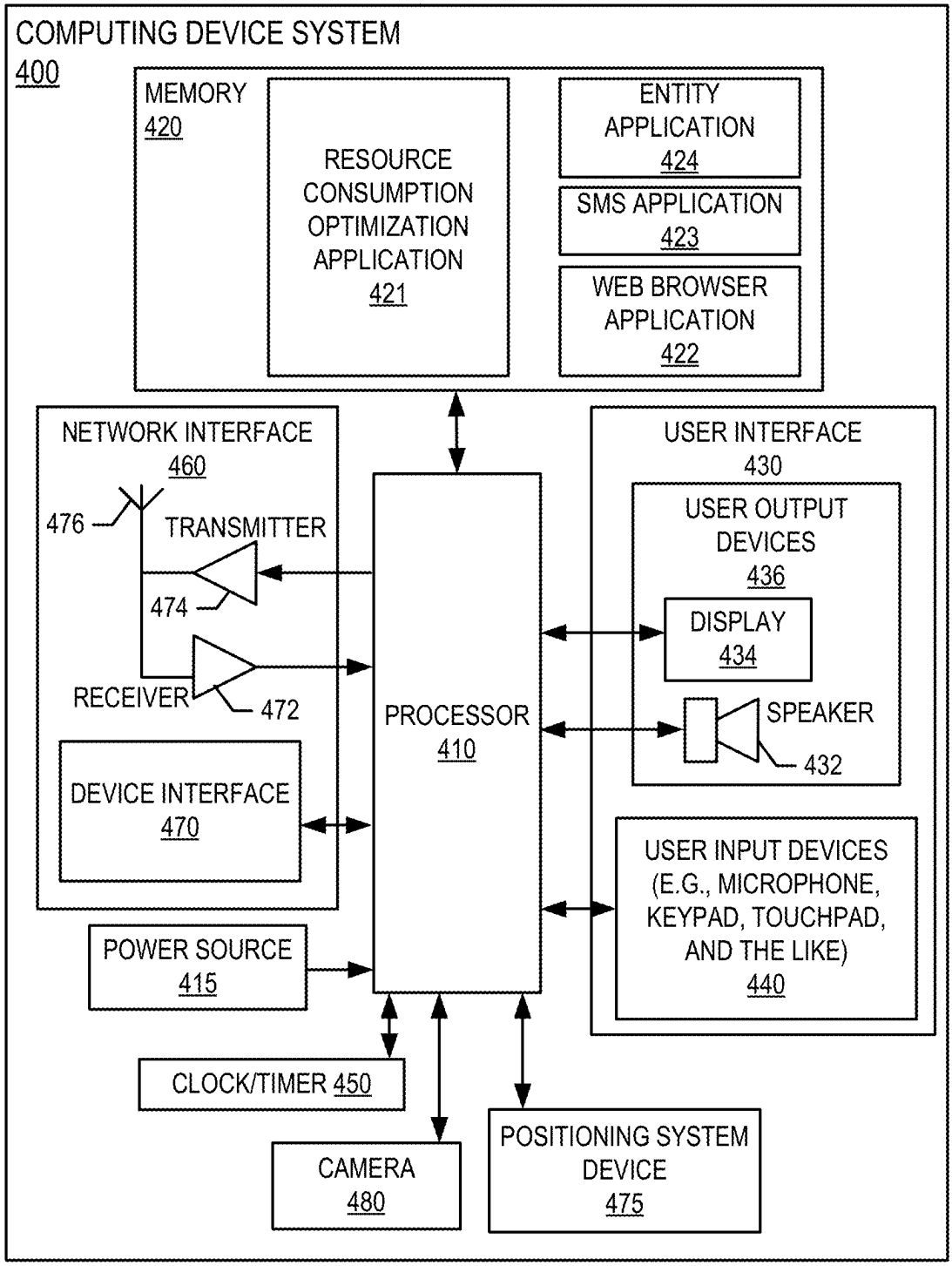

COMPUTING DEVICE SYSTEM
400
MEMORY
420
RESOURCE CONSUMPTION OPTIMIZATION APPLICATION
421
ENTITY APPLICATION
424
SMS APPLICATION
423
WEB BROWSER APPLICATION
422
NETWORK INTERFACE
460
476
TRANSMITTER
474
RECEIVER
472
DEVICE INTERFACE
470
PROCESSOR
410
USER INTERFACE
430
USER OUTPUT DEVICES
436
DISPLAY
434
SPEAKER
432
USER INPUT DEVICES (E.G., MICROPHONE, KEYPAD, TOUCHPAD, AND THE LIKE)
440
POWER SOURCE
415
CLOCK/TIMER 450
CAMERA
480
POSITIONING SYSTEM DEVICE
475

FIG. 4

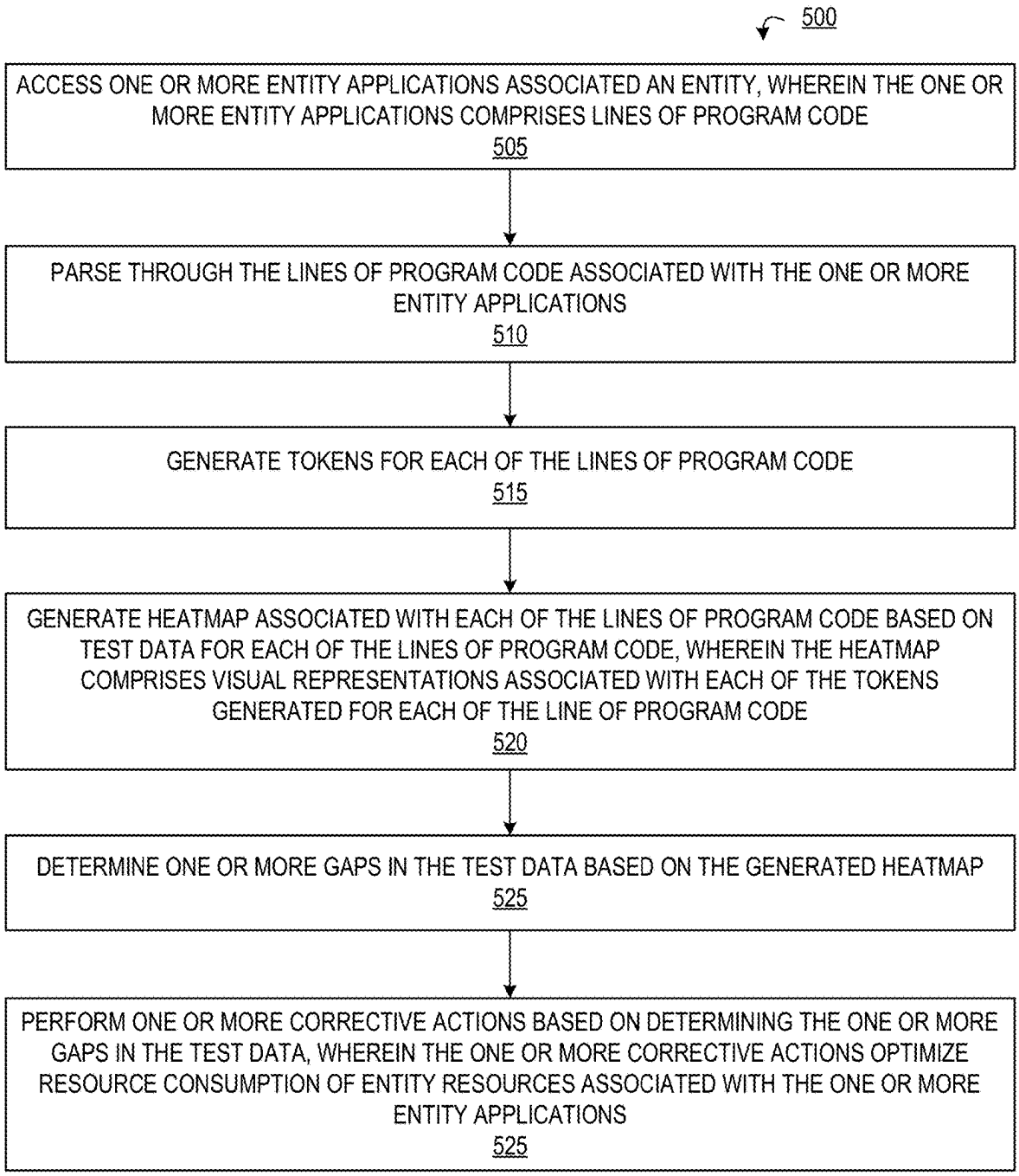

500
ACCESS ONE OR MORE ENTITY APPLICATIONS ASSOCIATED AN ENTITY, WHEREIN THE ONE OR MORE ENTITY APPLICATIONS COMPRISES LINES OF PROGRAM CODE
505
PARSE THROUGH THE LINES OF PROGRAM CODE ASSOCIATED WITH THE ONE OR MORE ENTITY APPLICATIONS
510
GENERATE TOKENS FOR EACH OF THE LINES OF PROGRAM CODE
515
GENERATE HEATMAP ASSOCIATED WITH EACH OF THE LINES OF PROGRAM CODE BASED ON TEST DATA FOR EACH OF THE LINES OF PROGRAM CODE, WHEREIN THE HEATMAP COMPRISES VISUAL REPRESENTATIONS ASSOCIATED WITH EACH OF THE TOKENS GENERATED FOR EACH OF THE LINE OF PROGRAM CODE
520
DETERMINE ONE OR MORE GAPS IN THE TEST DATA BASED ON THE GENERATED HEATMAP
525
PERFORM ONE OR MORE CORRECTIVE ACTIONS BASED ON DETERMINING THE ONE OR MORE GAPS IN THE TEST DATA, WHEREIN THE ONE OR MORE CORRECTIVE ACTIONS OPTIMIZE RESOURCE CONSUMPTION OF ENTITY RESOURCES ASSOCIATED WITH THE ONE OR MORE ENTITY APPLICATIONS
525

FIG. 5

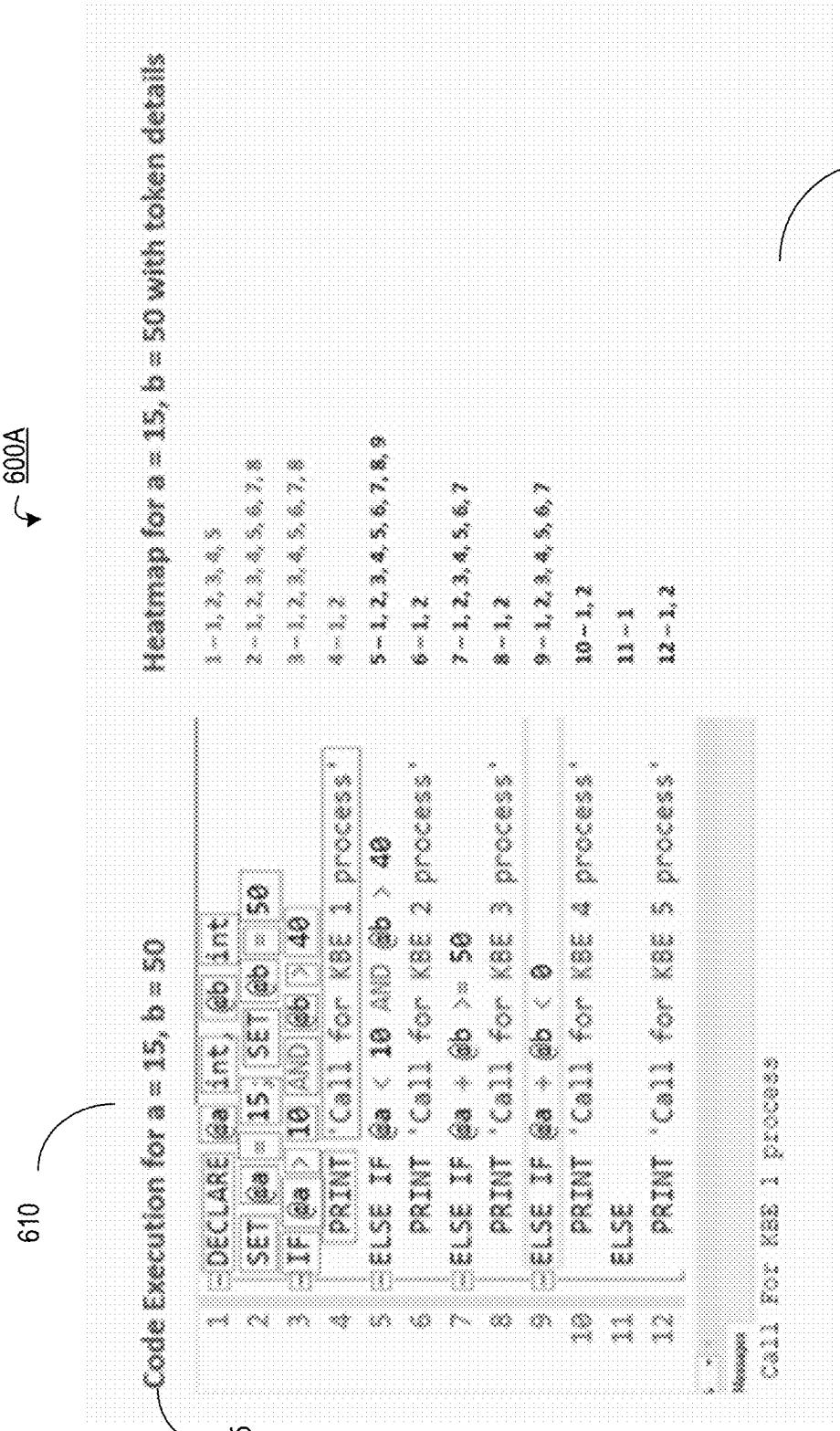

600A
610
615
650
Code Execution for a = 15, b = 50
Heatmap for a = 15, b = 50 with token details
1 -DECLARE @a int, @b int
2 SET @a = 15, SET @b = 50
3 -IF @a > 10 AND @b > 40
4 PRINT 'Call for KBE 1 process'
5 -ELSE IF @a < 10 AND @b > 40
6 PRINT 'Call for KBE 2 process'
7 -ELSE IF @a + @b >= 50
8 PRINT 'Call for KBE 3 process'
9 -ELSE IF @a + @b < 0
10 PRINT 'Call for KBE 4 process'
11 ELSE
12 PRINT 'Call for KBE 5 process'
1 - 1, 2, 3, 4, 5
2 - 1, 2, 3, 4, 5, 6, 7, 8
3 - 1, 2, 3, 4, 5, 6, 7, 8
4 - 1, 2
5 - 1, 2, 3, 4, 5, 6, 7, 8, 9
6 - 1, 2
7 - 1, 2, 3, 4, 5, 6, 7
8 - 1, 2
9 - 1, 2, 3, 4, 5, 6, 7
10 - 1, 2
11 - 1
12 - 1, 2
Call For KBE 1 process

FIG. 6A

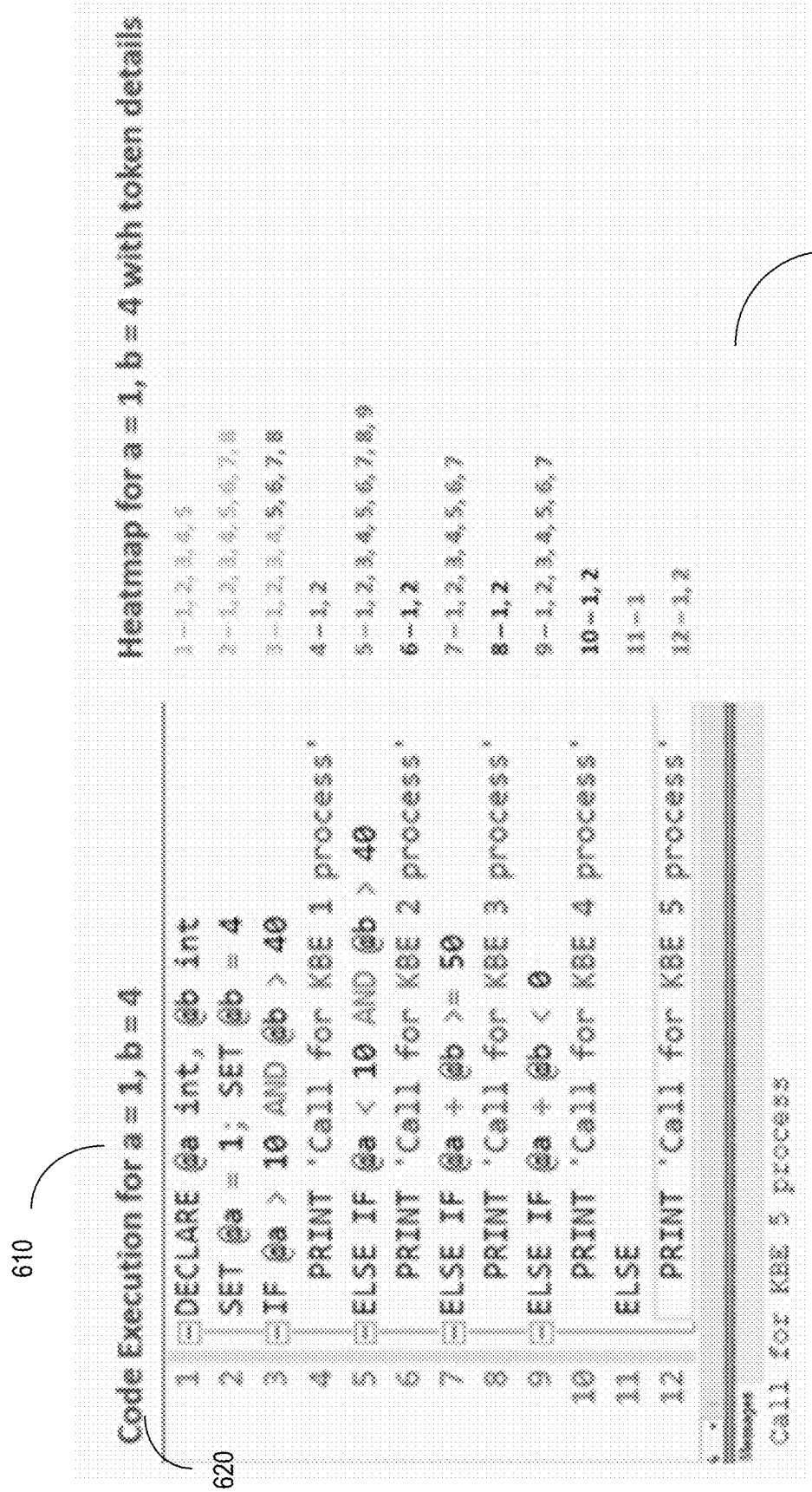

600B
610
620
655
Code Execution for a = 1, b = 4
Heatmap for a = 1, b = 4 with token details
1 DECLARE @a int, @b int
2 SET @a = 1; SET @b = 4
3 IF @a > 10 AND @b > 40
4 PRINT 'Call for KBE 1 process'
5 ELSE IF @a < 10 AND @b > 40
6 PRINT 'Call for KBE 2 process'
7 ELSE IF @a + @b >= 50
8 PRINT 'Call for KBE 3 process'
9 ELSE IF @a + @b < 0
10 PRINT 'Call for KBE 4 process'
11 ELSE
12 PRINT 'Call for KBE 5 process'
Messages
Call for KBE 5 process
1 – 1, 2, 3, 4, 5, 6
2 – 1, 2, 3, 4, 5, 6, 7, 8
3 – 1, 2, 3, 4, 5, 6, 7, 8
4 – 1, 2
5 – 1, 2, 3, 4, 5, 6, 7, 8, 9
6 – 1, 2
7 – 1, 2, 3, 4, 5, 6, 7
8 – 1, 2
9 – 1, 2, 3, 4, 5, 6, 7
10 – 1, 2
11 – 1
12 – 1, 2

FIG. 6B

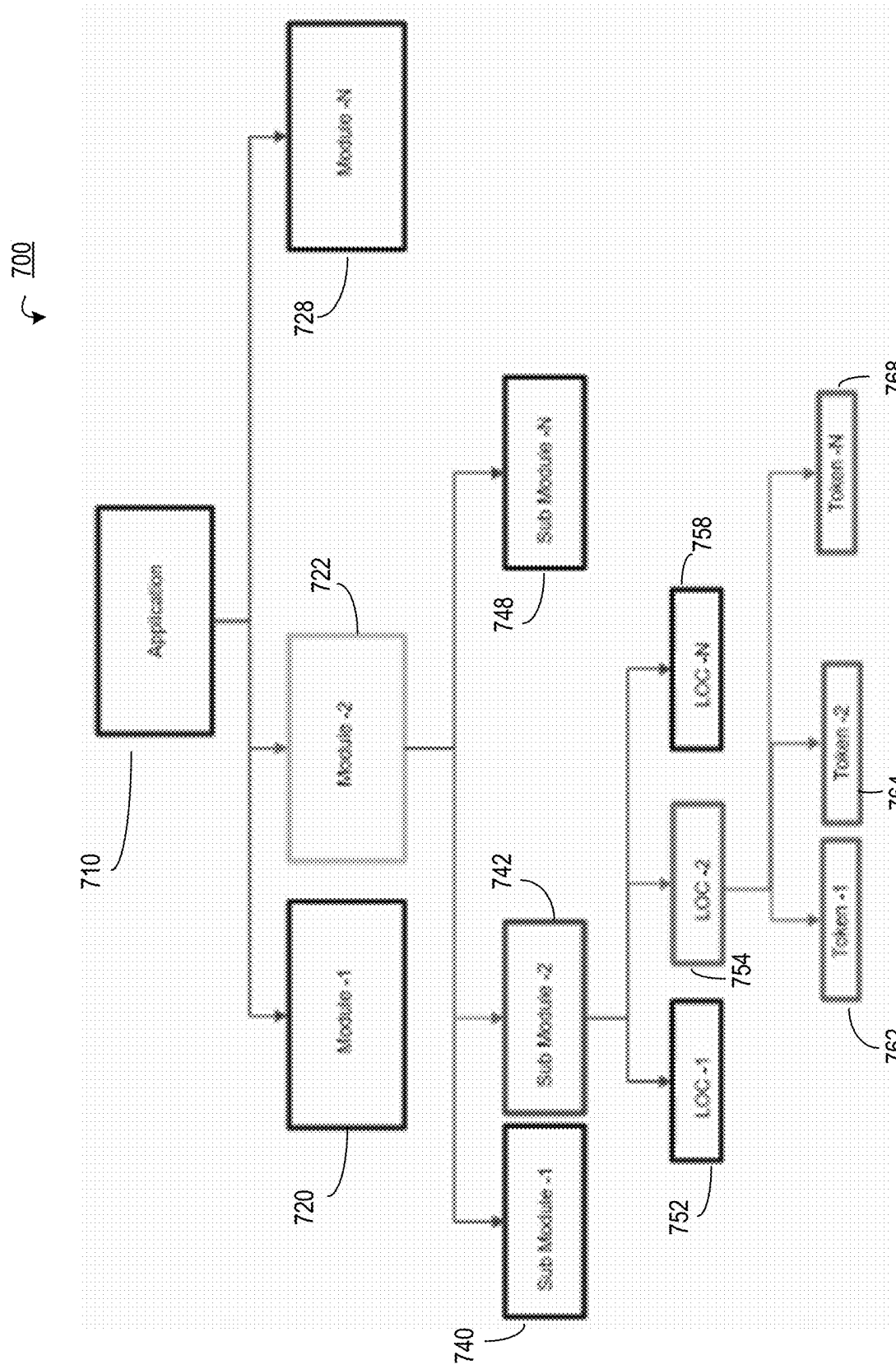

700
Application
710
Module -1
720
Module -2
722
Module -N
728
Sub Module -1
740
Sub Module -2
742
Sub Module -N
748
LOC -1
752
LOC -2
754
LOC -N
758
Token -1
762
Token -2
764
Token -N
768

FIG. 7

SYSTEM AND METHOD FOR GENERATING SOFTWARE CODE CRITICALITY AND EXECUTION HEATMAPS TO OPTIMIZE RESOURCE CONSUMPTION

BACKGROUND

[0001] There exists a need for a system for generating software code criticality and execution heatmaps to optimize resource consumption.

BRIEF SUMMARY

[0002] Embodiments of the present invention address the above needs and/or achieve other advantages by providing apparatuses (e.g., a system, computer program product and/or other devices) and methods for generating software code criticality and execution heatmaps to optimize resource consumption. The system embodiments may comprise one or more memory devices having computer readable program code stored thereon, a communication device, and one or more processing devices operatively coupled to the one or more memory devices, wherein the one or more processing devices are configured to execute the computer readable program code to carry out the invention. In computer program product embodiments of the invention, the computer program product comprises at least one non-transitory computer readable medium comprising computer readable instructions for carrying out the invention. Computer implemented method embodiments of the invention may comprise providing a computing system comprising a computer processing device and a non-transitory computer readable medium, where the computer readable medium comprises configured computer program instruction code, such that when said instruction code is operated by said computer processing device, said computer processing device performs certain operations to carry out the invention.

[0003] In some embodiments, the present invention accesses one or more entity applications associated an entity, wherein the one or more entity applications comprises lines of program code, parses through the lines of program code associated with the one or more entity applications, generates tokens for each of the lines of program code, generates heatmap associated with each of the lines of program code, wherein the heatmap comprises visual representations associated with each of the tokens generated for each of the line of program code, and displays the heatmap to one or more users.

[0004] In some embodiments, the present invention generates the heatmap associated with each of the lines of program code based on executing the lines of program code based on test data.

[0005] In some embodiments, the present invention determines one or more gaps in the test data based on the heatmap and performs one or more corrective actions based on determining the one or more gaps.

[0006] In some embodiments, the one or more corrective actions comprise at least one of generating and transmitting alerts to one or more users associated with the one or more entity applications, modifying at least one of the lines of program code, commenting out the at least one of the lines of program code, and rearranging the lines of program code.

[0007] In some embodiments, the one or more corrective actions improve the resource consumption of one or more entity systems executing the lines of program code associated with one or more entity applications.

[0008] In some embodiments, the present invention the visual representations comprise representations that denote criticality and execution frequency associated with each of the lines of program code.

[0009] In some embodiments, the present invention the one or more entity applications comprise modules, wherein each of the modules comprise submodules, and each of the submodules comprise the lines of program code.

[0010] The features, functions, and advantages that have been discussed may be achieved independently in various embodiments of the present invention or may be combined with yet other embodiments, further details of which can be seen with reference to the following description and drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] Having thus described embodiments of the invention in general terms, reference will now be made the accompanying drawings, wherein:

[0012] FIG. **1** provides a block diagram illustrating a system environment for generating software code criticality and execution heatmaps to optimize resource consumption, in accordance with an embodiment of the invention;

[0013] FIG. **2** provides a block diagram illustrating the entity system **200** of FIG. **1**, in accordance with an embodiment of the invention;

[0014] FIG. **3** provides a block diagram illustrating an resource consumption optimization system **300** of FIG. **1**, in accordance with an embodiment of the invention;

[0015] FIG. **4** provides a block diagram illustrating the computing device system **400** of FIG. **1**, in accordance with an embodiment of the invention;

[0016] FIG. **5** provides a process flow for generating software code criticality and execution heatmaps to optimize resource consumption, in accordance with an embodiment of the invention;

[0017] FIGS. **6**A and **6**B illustrate examples for generating software code criticality and execution heatmaps with test data, in accordance with an embodiment of the invention; and

[0018] FIG. **7** illustrates a heatmap generated by the resource consumption optimization system **300**, in accordance with an embodiment of the invention.

DETAILED DESCRIPTION OF EMBODIMENTS OF THE INVENTION

[0019] Embodiments of the present invention will now be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all, embodiments of the invention are shown. Indeed, the invention may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. Where possible, any terms expressed in the singular form herein are meant to also include the plural form and vice versa, unless explicitly stated otherwise. Also, as used herein, the term "a" and/or "an" shall mean "one or more," even though the phrase "one or more" is also used herein. Furthermore, when it is said herein that something is "based on" something else, it may be based on one or more other things as well. In other

words, unless expressly indicated otherwise, as used herein "based on" means "based at least in part on" or "based at least partially on." Like numbers refer to like elements throughout.

[0020] As described herein, the term "entity" may be any organization that creates, manages, develops, provides, maintains, and/or uses one or more applications (e.g., web applications, mobile applications, or the like) to perform one or more activities. In some embodiments, the entity may be a financial institution which may include any financial institutions such as commercial banks, thrifts, federal and state savings banks, savings and loan associations, credit unions, investment companies, insurance companies and the like. In some embodiments, the entity may be a non-financial institution.

[0021] Many of the example embodiments and implementations described herein contemplate interactions engaged in by a user with a computing device and/or one or more communication devices and/or secondary communication devices. A "user", as referenced herein, may refer to an entity or individual that has the ability and/or authorization to access and use one or more applications, systems, servers, and/or devices provided by the entity and/or the system of the present invention. Furthermore, as used herein, the term "user computing device" or "mobile device" may refer to mobile phones, computing devices, tablet computers, wearable devices, smart devices and/or any portable electronic device capable of receiving and/or storing data therein.

[0022] A "user interface" is any device or software that allows a user to input information, such as commands or data, into a device, or that allows the device to output information to the user. For example, the user interface includes a graphical user interface (GUI) or an interface to input computer-executable instructions that direct a processing device to carry out specific functions. The user interface typically employs certain input and output devices to input data received from a user or to output data to a user. These input and output devices may include a display, mouse, keyboard, button, touchpad, touch screen, microphone, speaker, LED, light, joystick, switch, buzzer, bell, and/or other user input/output device for communicating with one or more users.

[0023] Typically, an entity may develop, use, maintain, and/or the one or more entity applications. Such entity applications developed by one or more users may have different execution times, which may cause performance issues in entity servers hosting the entity applications. As such, there exists a need for a system to optimize the entity applications such that execution of the entity applications does not deteriorate the performance of the entity servers. The system of the invention provides a solution to this problem as explained below.

[0024] FIG. **1** provides a block diagram illustrating a system environment **100** for generating software code criticality and execution heatmaps to optimize resource consumption, in accordance with an embodiment of the invention. As illustrated in FIG. **1**, the environment **100** includes a resource consumption optimization system **300**, entity system **200**, and a computing device system **400**. One or more users **110** may be included in the system environment **100**, where the users **110** interact with the other entities of the system environment **100** via a user interface of the computing device system **400**. In some embodiments, the one or more user(s) **110** of the system environment **100** may be employees of an entity associated with the entity system **200** (e.g., software engineer, application developer, application tester, and/or the like). In some embodiments, the one or more user(s) **110** of the system environment **100** may further comprise end-users which may include, but are not limited to, customers, potential customers, or the like of the entity associated with the entity system **200**.

[0025] The entity system(s) **200** may be any system owned or otherwise controlled by an entity to support or perform one or more process steps described herein. In some embodiments, the entity is a financial institution. In some embodiments, the entity is a non-financial institution.

[0026] The resource consumption optimization system **300** is a system of the present invention for performing one or more process steps described herein. In some embodiments, the resource consumption optimization system **300** may be an independent system. In some embodiments, the resource consumption optimization system **300** may be a part of the entity system **200**.

[0027] The resource consumption optimization system **300**, the entity system **200**, and/or the computing device system **400** may be in network communication across the system environment **100** through the network **150**. The network **150** may include a local area network (LAN), a wide area network (WAN), and/or a global area network (GAN). The network **150** may provide for wireline, wireless, or a combination of wireline and wireless communication between devices in the network. In one embodiment, the network **150** includes the Internet. In general, the resource consumption optimization system **300** is configured to communicate information or instructions with the entity system **200**, and/or the computing device system **400** across the network **150**.

[0028] The computing device system **400** may be a computing device of the user **110**. In general, the computing device system **400** communicates with the user **110** via a user interface of the computing device system **400**, and in turn is configured to communicate information or instructions with the resource consumption optimization system **300** and/or entity system **200** across the network **150**.

[0029] FIG. **2** provides a block diagram illustrating the entity system **200**, in greater detail, in accordance with embodiments of the invention. As illustrated in FIG. **2**, in one embodiment of the invention, the entity system **200** includes one or more processing devices **220** operatively coupled to a network communication interface **210** and a memory device **230**. In certain embodiments, the entity system **200** is operated by an entity, such as a financial institution, while in other embodiments, the entity system **200** is operated by an entity other than a financial institution.

[0030] It should be understood that the memory device **230** may include one or more databases or other data structures/repositories. The memory device **230** also includes computer-executable program code that instructs the processing device **220** to operate the network communication interface **210** to perform certain communication functions of the entity system **200** described herein. For example, in one embodiment of the entity system **200**, the memory device **230** includes, but is not limited to, a network server application **240**, a resource consumption optimization application **250**, one or more entity applications **260**, and a data repository **280**. The computer-executable program code of the network server application **240**, the resource consumption optimization application **250**, and the one or more

entity applications **260** to perform certain logic, data-extraction, and data-storing functions of the entity system **200** described herein, as well as communication functions of the entity system **200**.

[0031] The network server application **240**, the resource consumption optimization application **250**, and the one or more entity applications **260** are configured to store data in the data repository **280** or to use the data stored in the data repository **280** when communicating through the network communication interface **210** with the resource consumption optimization system **300**, and the computing device system **400** to perform one or more process steps described herein. In some embodiments, the entity system **200** may receive instructions from the resource consumption optimization system **300** via the resource consumption optimization application **250** to perform certain operations. The resource consumption optimization application **250** may be provided by the resource consumption optimization system **300**.

[0032] FIG. **3** provides a block diagram illustrating the resource consumption optimization system **300** in greater detail, in accordance with embodiments of the invention. As illustrated in FIG. **3**, in one embodiment of the invention, the resource consumption optimization system **300** includes one or more processing devices **320** operatively coupled to a network communication interface **310** and a memory device **330**. In certain embodiments, the resource consumption optimization system **300** is operated by an entity, such as a financial institution, while in other embodiments, the resource consumption optimization system **300** is operated by an entity other than a financial institution. In some embodiments, the resource consumption optimization system **300** is owned or operated by the entity of the entity system **200**. In some embodiments, the resource consumption optimization system **300** may be an independent system. In alternate embodiments, the resource consumption optimization system **300** may be a part of the entity system **200**.

[0033] It should be understood that the memory device **330** may include one or more databases or other data structures/repositories. The memory device **330** also includes computer-executable program code that instructs the processing device **320** to operate the network communication interface **310** to perform certain communication functions of the resource consumption optimization system **300** described herein. For example, in one embodiment of the resource consumption optimization system **300**, the memory device **330** includes, but is not limited to, a network provisioning application **340**, a parsing application **350**, a token generation application **355**, an execution application **360**, a criticality determination application **365**, a heatmap generation application **370**, an alert transmission application **375**, a mitigation application **380**, and a data repository **390** comprising data processed or accessed by one or more applications in the memory device **330**. The computer-executable program code of the network provisioning application **340**, the parsing application **350**, the token generation application **355**, the execution application **360**, the criticality determination application **365**, the heatmap generation application **370**, the alert transmission application **375**, and the mitigation application **380** may instruct the processing device **320** to perform certain logic, data-processing, and data-storing functions of the resource consumption optimization system **300** described herein, as well as communication functions of the resource consumption optimization system **300**.

[0034] The network provisioning application **340**, the parsing application **350**, the token generation application **355**, the execution application **360**, the criticality determination application **365**, the heatmap generation application **370**, the alert transmission application **375**, and the mitigation application **380** are configured to invoke or use the data in the data repository **390** when communicating through the network communication interface **310** with the entity system **200**, and the computing device system **400**. In some embodiments, the network provisioning application **340**, the parsing application **350**, the token generation application **355**, the execution application **360**, the criticality determination application **365**, the heatmap generation application **370**, the alert transmission application **375**, and the mitigation application **380** may store the data extracted or received from the entity system **200** and the computing device system **400** in the data repository **390**. In some embodiments, the network provisioning application **340**, the parsing application **350**, the token generation application **355**, the execution application **360**, the criticality determination application **365**, the heatmap generation application **370**, the alert transmission application **375**, and the mitigation application **380** may be a part of a single application. One or more processes performed by the network provisioning application **340**, the parsing application **350**, the token generation application **355**, the execution application **360**, the criticality determination application **365**, the heatmap generation application **370**, the alert transmission application **375**, and the mitigation application **380** are described in detail below.

[0035] FIG. **4** provides a block diagram illustrating a computing device system **400** of FIG. **1** in more detail, in accordance with embodiments of the invention. However, it should be understood that the computing device system **400** is merely illustrative of one type of computing device system that may benefit from, employ, or otherwise be involved with embodiments of the present invention and, therefore, should not be taken to limit the scope of embodiments of the present invention. The computing devices may include any one of portable digital assistants (PDAs), pagers, mobile televisions, mobile phone, entertainment devices, desktop computers, workstations, laptop computers, cameras, video recorders, audio/video player, radio, GPS devices, wearable devices, Internet-of-things devices, augmented reality devices, virtual reality devices, automated teller machine devices, electronic kiosk devices, or any combination of the aforementioned.

[0036] Some embodiments of the computing device system **400** include a processor **410** communicably coupled to such devices as a memory **420**, user output devices **436**, user input devices **440**, a network interface **460**, a power source **415**, a clock or other timer **450**, a camera **480**, and a positioning system device **475**. The processor **410**, and other processors described herein, generally include circuitry for implementing communication and/or logic functions of the computing device system **400**. For example, the processor **410** may include a digital signal processor device, a microprocessor device, and various analog to digital converters, digital to analog converters, and/or other support circuits. Control and signal processing functions of the computing device system **400** are allocated between these devices according to their respective capabilities. The processor **410**

thus may also include the functionality to encode and interleave messages and data prior to modulation and transmission. The processor **410** can additionally include an internal data modem. Further, the processor **410** may include functionality to operate one or more software programs, which may be stored in the memory **420**. For example, the processor **410** may be capable of operating a connectivity program, such as a web browser application **422**. The web browser application **422** may then allow the computing device system **400** to transmit and receive web content, such as, for example, location-based content and/or other web page content, according to a Wireless Application Protocol (WAP), Hypertext Transfer Protocol (HTTP), and/or the like.

[0037] The processor **410** is configured to use the network interface **460** to communicate with one or more other devices on the network **150**. In this regard, the network interface **460** includes an antenna **476** operatively coupled to a transmitter **474** and a receiver **472** (together a "transceiver"). The processor **410** is configured to provide signals to and receive signals from the transmitter **474** and receiver **472**, respectively. The signals may include signaling information in accordance with the air interface standard of the applicable cellular system of the wireless network **150**. In this regard, the computing device system **400** may be configured to operate with one or more air interface standards, communication protocols, modulation types, and access types. By way of illustration, the computing device system **400** may be configured to operate in accordance with any of a number of first, second, third, and/or fourth-generation communication protocols and/or the like. For example, the computing device system **400** may be configured to operate in accordance with second-generation (2G) wireless communication protocols IS-136 (time division multiple access (TDMA)), GSM (global system for mobile communication), and/or IS-95 (code division multiple access (CDMA)), or with third-generation (3G) wireless communication protocols, such as Universal Mobile Telecommunications System (UMTS), CDMA2000, wideband CDMA (WCDMA) and/or time division-synchronous CDMA (TD-SCDMA), with fourth-generation (4G) wireless communication protocols, with LTE protocols, with 4GPP protocols and/or the like. The computing device system **400** may also be configured to operate in accordance with non-cellular communication mechanisms, such as via a wireless local area network (WLAN) or other communication/data networks.

[0038] As described above, the computing device system **400** has a user interface that is, like other user interfaces described herein, made up of user output devices **436** and/or user input devices **440**. The user output devices **436** include a display **430** (e.g., a liquid crystal display or the like) and a speaker **432** or other audio device, which are operatively coupled to the processor **410**.

[0039] The user input devices **440**, which allow the computing device system **400** to receive data from a user such as the user **110** may include any of a number of devices allowing the computing device system **400** to receive data from the user **110**, such as a keypad, keyboard, touch-screen, touchpad, microphone, mouse, joystick, other pointer device, button, soft key, and/or other input device(s). The user interface may also include a camera **480**, such as a digital camera.

[0040] The computing device system **400** may also include a positioning system device **475** that is configured to be used by a positioning system to determine a location of the computing device system **400**. For example, the positioning system device **475** may include a GPS transceiver. In some embodiments, the positioning system device **475** is at least partially made up of the antenna **476**, transmitter **474**, and receiver **472** described above. For example, in one embodiment, triangulation of cellular signals may be used to identify the approximate or exact geographical location of the computing device system **400**. In other embodiments, the positioning system device **475** includes a proximity sensor or transmitter, such as an RFID tag, that can sense or be sensed by devices known to be located proximate a merchant or other location to determine that the computing device system **400** is located proximate these known devices.

[0041] The computing device system **400** further includes a power source **415**, such as a battery, for powering various circuits and other devices that are used to operate the computing device system **400**. Embodiments of the computing device system **400** may also include a clock or other timer **450** configured to determine and, in some cases, communicate actual or relative time to the processor **410** or one or more other devices.

[0042] The computing device system **400** also includes a memory **420** operatively coupled to the processor **410**. As used herein, memory includes any computer readable medium (as defined herein below) configured to store data, code, or other information. The memory **420** may include volatile memory, such as volatile Random Access Memory (RAM) including a cache area for the temporary storage of data. The memory **420** may also include non-volatile memory, which can be embedded and/or may be removable. The non-volatile memory can additionally or alternatively include an electrically erasable programmable read-only memory (EEPROM), flash memory or the like.

[0043] The memory **420** can store any of a number of applications which comprise computer-executable instructions/code executed by the processor **410** to implement the functions of the computing device system **400** and/or one or more of the process/method steps described herein. For example, the memory **420** may include such applications as a conventional web browser application **422**, a resource consumption optimization application **421**, an entity application **424**, or the like. These applications also typically instructions to a graphical user interface (GUI) on the display **430** that allows the user **110** to interact with the entity system **200**, the resource consumption optimization system **300**, and/or other devices or systems. The memory **420** of the computing device system **400** may comprise a Short Message Service (SMS) application **423** configured to send, receive, and store data, information, communications, alerts, and the like via the wireless network **150**.

[0044] The memory **420** can also store any of a number of pieces of information, and data, used by the computing device system **400** and the applications and devices that make up the computing device system **400** or are in communication with the computing device system **400** to implement the functions of the computing device system **400** and/or the other systems described herein.

[0045] FIG. **5** provides a process flow for generating software code criticality and execution heatmaps to optimize resource consumption, in accordance with an embodiment

of the invention. As shown in block **505**, the system accesses one or more entity applications associated an entity, wherein the one or more entity applications comprises lines of program code. The one or more entity applications may be any applications that are being developed, maintained, utilized, managed, and/or the like by the entity. The one or more entity applications may be developed, tested and deployed into real-time environment by one or more users (e.g., application developers, application testers, software engineers, and/or the like via the computing device system **400**). The one or more entity applications may comprise one or more modules, wherein each of the one or more modules may comprise one or more sub-modules, and each of the one or more sub-modules may comprise the lines of program code. In some embodiments, the one or more modules may not have one or more sub-modules and may have the lines of program code directly in the one or more modules.

[0046] As shown in block **510**, the system parses through the lines of program code associated with the one or more entity applications. In some embodiments, the system may parse through each of the modules, the submodules, and the lines of the program code. The system analyzes the lines of program code and may identify different parameters, variables, values, functions, methods, and/or the like in each of the lines of program code. As shown in block **515**, the system generates tokens for each of the lines of program code.

[0047] As shown in block **520**, the system generates heatmap associated with each of the lines of program code based on test data for each of the lines of program code, wherein the heatmap comprises visual representations associated with each of the tokens generated for each of the lines of program code. In some embodiments, the system may receive the test data from the one or more users. In some embodiments, the system may extract the test data from a data repository. In some embodiments, the system may dynamically formulate the test data for testing the lines of program code. In some such embodiments, the system may generate the test data based on historical test data, real-time data, and/or the like associated with the one or more applications associated with the lines of program code. The system may generate the heatmap based on the test data for each of the lines of program code. The system may execute each of the modules, submodules, and the lines of the program code using the test data to identify criticality and execution frequency associated with modules, submodules, and the lines of program code. Based on the execution, the system may generate the heatmap comprising visual representations for each of the tokens in the lines of program codes, modules, submodules, and/or the like. The visual representations in the heatmap denote the criticality and the execution frequency associated with the tokens, lines of code, modules, submodules, and/or the like. In some embodiments, the system may use any kind of visual representations for displaying the heatmap and the criticality and execution frequency in the heatmap.

[0048] As shown in block **525**, the system determines one or more gaps in the test data based on the generated heatmap. Based on the generated heatmap, the system may determine that a submodule of the 'n' submodules associated with a first module of the 'm' modules associated with an application is not executed at least once using the test data and may therefore determine a gap in the test data that resulted in non-execution of the submodule.

[0049] As shown in block **530**, the system performs one or more corrective actions based on determining the one or more gaps in the test data, wherein the one or more corrective actions optimize resource consumption of entity resources associated with the one or more entity applications. The one or more corrective actions may comprise generating and transmitting alerts to one or more users associated with the one or more entity applications, modifying at least one of the lines of program code, commenting out the at least one of the lines of program code, rearranging the lines of program code, and/or the like.

[0050] FIG. **6**A and **6**B illustrate examples for generating software code criticality and execution heatmaps with test data, in accordance with an embodiment of the invention. It should be understood that the examples described herein are for explanatory purposes only and the program code may be more complicated in real-time. As shown, program code **610** may be a part of a module and/or a submodule of an entity application, where the program code **610** comprises two sets of parameters that are defined based on test data or real-time data.

[0051] In FIG. **6**A, the parameters are set as shown in label in **615**, where 'a' is assigned a value of '15' and 'b' is assigned a value of '50.' As shown, a heatmap **650** is generated for the values corresponding to label **615**, where the heatmap **650** comprises visual representations associated with tokens associated with the lines of program code **610**. As shown, the system generates five tokens for first line in program code **610**, eight tokens for second line in program code **610**, and the like, where each of the tokens are represented via different visual representations (e.g., different color) based on the criticality and execution frequency of the tokens. As shown in heatmap **650**, the first four lines of the program code **610** are executed once and are therefore displayed using a first visual representation element (e.g., first color) and the remaining lines of the program code **610** are not executed at least once using the test data in label **615** and are therefore displayed using a second visual representation element (e.g., second color).

[0052] In FIG. **6**A, the parameters are set as shown in label in **620**, where 'a' is assigned a value of '1' and 'b' is assigned a value of '4.' As shown, a heatmap **655** is generated for the values corresponding to label **620**, where the heatmap **655** comprises visual representations associated with tokens associated with the lines of program code **610**. As shown, lines 1-5, 7, 9,11, and 12 are executed when the values in label **620** are provided to the program code **610**. As such, the lines that are executed are displayed using a first visual representation element (e.g., first color) and the remaining lines that are not executed are displayed using a second visual representation element (e.g., second color). In addition, each token in the line is visually represented using different visual representation elements (e.g., different colors, shading, or the like) based on the criticality and/or the execution frequency. For example, as shown in third line, the tokens 1-4 are displayed in a third color and tokens 5-8 are displayed in a fourth color.

[0053] FIG. **7** illustrates a heatmap generated by the resource consumption optimization system **300**, in accordance with an embodiment of the invention. As shown, the application **710** may comprise module '1' **720**, module '2' **722**, through module 'N' **728**. Module '2' **722** may comprise Submodule '1' **740**, Submodule '2' **742** through Submodule 'N' **748**. Submodule '2' **742** may further comprise functions

Log '1' **752**, Log '2' **754**, through Log 'N' **758**. Function Log '2' **754** may further comprise Token '1' **762**, Token '2' **764**, through Token 'N' **768**. The value of 'N' in each of these different elements may vary and is not constant throughout the application **710**. Upon execution, the system may determine that the Module '2' **722** is a critical module based at least on execution frequency (e.g., may determine that it is executed 'X' number of times) and may display it via a first visual representation element (e.g., first color). Upon execution, the system may determine that the Submodule '2' **742** is a critical submodule based at least on execution frequency (e.g., may determine that it sis executed 'Y' number of times) and may display it via a second visual representation element. The system may further determine that function Log '2' **754** is executed at least once (e.g., 'Z' number of times, where 'X' and 'Y' may be greater than 'Z') compared to the other functions and may represent the function Log '2' **754** using a third visual representation element (e.g., third color). In some embodiments, the first visual representation element, the second visual representation element, and the third visual element are different. In some embodiments, at least two of the first visual representation element, the second visual representation element, and the third visual element are the same.

[0054] As will be appreciated by one of skill in the art, the present invention may be embodied as a method (including, for example, a computer-implemented process, a business process, and/or any other process), apparatus (including, for example, a system, machine, device, computer program product, and/or the like), or a combination of the foregoing. Accordingly, embodiments of the present invention may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, and the like), or an embodiment combining software and hardware aspects that may generally be referred to herein as a "system." Furthermore, embodiments of the present invention may take the form of a computer program product on a computer-readable medium having computer-executable program code embodied in the medium.

[0055] Any suitable transitory or non-transitory computer readable medium may be utilized. The computer readable medium may be, for example but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, or device. More specific examples of the computer readable medium include, but are not limited to, the following: an electrical connection having one or more wires; a tangible storage medium such as a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a compact disc read-only memory (CD-ROM), or other optical or magnetic storage device.

[0056] In the context of this document, a computer readable medium may be any medium that can contain, store, communicate, or transport the program for use by or in connection with the instruction execution system, apparatus, or device. The computer usable program code may be transmitted using any appropriate medium, including but not limited to the Internet, wireline, optical fiber cable, radio frequency (RF) signals, or other mediums.

[0057] Computer-executable program code for carrying out operations of embodiments of the present invention may be written in an object oriented, scripted or unscripted programming language. However, the computer program code for carrying out operations of embodiments of the present invention may also be written in conventional procedural programming languages, such as the "C" programming language or similar programming languages.

[0058] Embodiments of the present invention are described above with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems), and computer program products. It will be understood that each block of the flowchart illustrations and/or block diagrams, and/or combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer-executable program code portions. These computer-executable program code portions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a particular machine, such that the code portions, which execute via the processor of the computer or other programmable data processing apparatus, create mechanisms for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

[0059] These computer-executable program code portions may also be stored in a computer-readable memory that can direct a computer or other programmable data processing apparatus to function in a particular manner, such that the code portions stored in the computer readable memory produce an article of manufacture including instruction mechanisms which implement the function/act specified in the flowchart and/or block diagram block(s).

[0060] The computer-executable program code may also be loaded onto a computer or other programmable data processing apparatus to cause a series of operational steps to be performed on the computer or other programmable apparatus to produce a computer-implemented process such that the code portions which execute on the computer or other programmable apparatus provide steps for implementing the functions/acts specified in the flowchart and/or block diagram block(s). Alternatively, computer program implemented steps or acts may be combined with operator or human implemented steps or acts in order to carry out an embodiment of the invention.

[0061] As the phrase is used herein, a processor may be "configured to" perform a certain function in a variety of ways, including, for example, by having one or more general-purpose circuits perform the function by executing particular computer-executable program code embodied in computer-readable medium, and/or by having one or more application-specific circuits perform the function.

[0062] Embodiments of the present invention are described above with reference to flowcharts and/or block diagrams. It will be understood that steps of the processes described herein may be performed in orders different than those illustrated in the flowcharts. In other words, the processes represented by the blocks of a flowchart may, in some embodiments, be in performed in an order other that the order illustrated, may be combined or divided, or may be performed simultaneously. It will also be understood that the blocks of the block diagrams illustrated, in some embodiments, merely conceptual delineations between systems and one or more of the systems illustrated by a block in the block diagrams may be combined or share hardware and/or software with another one or more of the systems illustrated by a block in the block diagrams. Likewise, a device, system, apparatus, and/or the like may be made up of one or more

devices, systems, apparatuses, and/or the like. For example, where a processor is illustrated or described herein, the processor may be made up of a plurality of microprocessors or other processing devices which may or may not be coupled to one another. Likewise, where a memory is illustrated or described herein, the memory may be made up of a plurality of memory devices which may or may not be coupled to one another.

[0063] While certain exemplary embodiments have been described and shown in the accompanying drawings, it is to be understood that such embodiments are merely illustrative of, and not restrictive on, the broad invention, and that this invention not be limited to the specific constructions and arrangements shown and described, since various other changes, combinations, omissions, modifications and substitutions, in addition to those set forth in the above paragraphs, are possible. Those skilled in the art will appreciate that various adaptations and modifications of the just described embodiments can be configured without departing from the scope and spirit of the invention. Therefore, it is to be understood that, within the scope of the appended claims, the invention may be practiced other than as specifically described herein.

Claims:

1. A system for generating software code criticality and execution heatmaps to optimize resource consumption, comprising:

at least one processing device;

at least one memory device; and

a module stored in the at least one memory device comprising executable instructions that when executed by the at least one processing device, cause the at least one processing device to:

access one or more entity applications associated an entity, wherein the one or more entity applications comprises lines of program code;

parse through the lines of program code associated with the one or more entity applications;

generate tokens for each of the lines of program code;

generate heatmap associated with each of the lines of program code, wherein the heatmap comprises visual representations associated with each of the tokens generated for each of the line of program code; and

display the heatmap to one or more users.

2. The system according to claim **1**, wherein the executable instructions cause the at least one processing device to generate the heatmap associated with each of the lines of program code based on executing the lines of program code based on test data.

3. The system according to claim **1**, wherein the executable instructions cause the at least one processing device to:

determine one or more gaps in the test data based on the heatmap; and

perform one or more corrective actions based on determining the one or more gaps.

4. The system according to claim **3**, wherein the one or more corrective actions comprise at least one of:

generating and transmitting alerts to one or more users associated with the one or more entity applications;

modifying at least one of the lines of program code;

commenting out the at least one of the lines of program code; and

rearranging the lines of program code.

5. The system according to claim **4**, wherein the one or more corrective actions improve resource consumption of one or more entity systems executing the lines of program code associated with one or more entity applications.

6. The system according to claim **1**, wherein the visual representations comprise representations that denote criticality and execution frequency associated with each of the lines of program code.

7. The system according to claim **1**, wherein the one or more entity applications comprise modules, wherein each of the modules comprise submodules, and each of the submodules comprise the lines of program code.

8. A computer program product for generating software code criticality and execution heatmaps to optimize resource consumption, comprising a non-transitory computer-readable storage medium having computer-executable instructions for:

accessing one or more entity applications associated an entity, wherein the one or more entity applications comprises lines of program code;

parsing through the lines of program code associated with the one or more entity applications;

generating tokens for each of the lines of program code;

generating heatmap associated with each of the lines of program code, wherein the heatmap comprises visual representations associated with each of the tokens generated for each of the line of program code; and

displaying the heatmap to one or more users.

9. The computer program product according to claim **8**, wherein the non-transitory computer-readable storage medium comprises computer-executable instructions for generating the heatmap associated with each of the lines of program code based on executing the lines of program code based on test data.

10. The computer program product according to claim **9**, wherein the non-transitory computer-readable storage medium comprises computer-executable instructions for:

determining one or more gaps in the test data based on the heatmap; and

performing one or more corrective actions based on determining the one or more gaps.

11. The computer program product according to claim **10**, wherein the one or more corrective actions comprise at least one of:

generating and transmitting alerts to one or more users associated with the one or more entity applications;

modifying at least one of the lines of program code;

commenting out the at least one of the lines of program code; and

rearranging the lines of program code.

12. The computer program product according to claim **11**, wherein the one or more corrective actions improve resource consumption of one or more entity systems executing the lines of program code associated with one or more entity applications.

13. The computer program product according to claim **8**, wherein the visual representations comprise representations that denote criticality and execution frequency associated with each of the lines of program code.

14. The computer program product according to claim **8**, wherein the one or more entity applications comprise modules, wherein each of the modules comprise submodules, and each of the submodules comprise the lines of program code.

15. A computerized method for generating software code criticality and execution heatmaps to optimize resource consumption, the method comprising:

accessing one or more entity applications associated an entity, wherein the one or more entity applications comprises lines of program code;

parsing through the lines of program code associated with the one or more entity applications;

generating tokens for each of the lines of program code;

generating heatmap associated with each of the lines of program code, wherein the heatmap comprises visual representations associated with each of the tokens generated for each of the line of program code; and

displaying the heatmap to one or more users.

16. The computerized method according to claim **15**, wherein the method comprises generating the heatmap associated with each of the lines of program code based on executing the lines of program code based on test data.

17. The computerized method according to claim **16**, wherein the method further comprises:

determining one or more gaps in the test data based on the heatmap; and

performing one or more corrective actions based on determining the one or more gaps.

18. The computerized method according to claim **17**, wherein the one or more corrective actions comprise at least one of:

generating and transmitting alerts to one or more users associated with the one or more entity applications;

modifying at least one of the lines of program code;

commenting out the at least one of the lines of program code; and

rearranging the lines of program code.

19. The computerized method according to claim **15**, wherein the visual representations comprise representations that denote criticality and execution frequency associated with each of the lines of program code.

20. The computerized method according to claim **15**, wherein the one or more entity applications comprise modules, wherein each of the modules comprise submodules, and each of the submodules comprise the lines of program code.

* * * * *